

**Legend:**

- eQTL** (Blue pentagon)
- Gene** (Grey rectangle)
- Region 1\*** (Orange diamond)
- Region 2\*** (Purple diamond)

**Logic:**

- Red line: "No"
- Green line: "Yes"

**\*region 1 and region 2 are homeologous region (duplicates)**

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graph TD
    A[eQTL in Region 1] -- Green --> B[Gene in Region 2]
    A -- Red --> C[eQTL in Region 2]
    B -- Green --> D[Inter-chromosomal ohnolog]
    B -- Red --> E[Gene in same chr as eQTL]
    C -- Green --> F[Gene in Region 1]
    C -- Red --> G[Gene in same chr as eQTL]
    E -- Red --> H[eQTL is present]
    E -- Green --> I[Gene in Region 1]
    G -- Red --> J[eQTL is present]
    G -- Green --> K[Gene in Region 2]
    I -- Green --> L[Intra-chromosomal syntenic]
    I -- Red --> M[Intra-chromosomal non syntenic]
    K -- Green --> N[Intra-chromosomal syntenic]
    K -- Red --> O[Intra-chromosomal non syntenic]
```

**Flowchart Details:**

- Start:** eQTL in Region 1
  - Yes (Green):** Gene in Region 2
    - Yes (Green):** Inter-chromosomal ohnolog
    - No (Red):** Gene in same chr as eQTL
      - No (Red):** eQTL is present
      - Yes (Green):** Gene in Region 1
        - Yes (Green):** Intra-chromosomal syntenic
        - No (Red):** Intra-chromosomal non syntenic
  - No (Red):** eQTL in Region 2
    - Yes (Green):** Gene in Region 1
      - Yes (Green):** Inter-chromosomal ohnolog
      - No (Red):** Gene in same chr as eQTL
        - No (Red):** eQTL is present
        - Yes (Green):** Gene in Region 2
          - Yes (Green):** Intra-chromosomal syntenic
          - No (Red):** Intra-chromosomal non syntenic
    - No (Red):** Gene in Region 2
      - Yes (Green):** Gene is present
      - No (Red):** NA